Phys. 2110 Solution HW # 06

6. 55, 57, 68, 82, 93, 96, 101

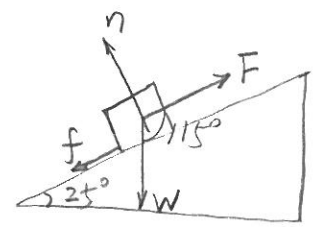
6.55 $K = \int \frac{1}{2}(dm)v^2$. $dm = \frac{M}{L}dx$, so

$$K = \int_0^L \frac{1}{2}\left(\frac{M}{L}dx\right)\left(\frac{2\pi x}{T}\right)^2 = \frac{1}{2}\left(\frac{M}{L}\right)\left(\frac{4\pi^2}{T^2}\right)\int_0^L x^2\,dx = \frac{1}{2}\left(\frac{M}{L}\right)\left(\frac{4\pi^2}{T^2}\right)\left(\frac{L^3}{3}\right) = \frac{2}{3}\pi^2 M\frac{L^2}{T^2}$$

There are 5 revolutions in 3 seconds, so $T = \frac{3}{5}s = 0.60s$

$$K = \frac{2}{3}\pi^2(12.0kg)(2.00m)^2/(0.60s)^2 = 877J$$

6.57 (a) $W_F = (F\cos\phi)s$



Both $\vec{F}$ and $\vec{s}$ are parallel to the incline and in the same direction, so $\phi = 90°$

$$W_F = Fs = (140N)(3.80m) = 532J$$

(b) The direction of the displacement and of the gravity force are shown in the Figure. $W_w = (W\cos\phi)s$

$\phi = 115°$, so $W_w = (196N)(\cos 115°)(3.80m) = -315J$

(c) The normal force is perpendicular to the displacement ($\phi = 90°$), so $W_n = 0$

(d) $n = W\cos 25°$, so $f_k = \mu_k n = \mu_k n\cos 25° = (0.30)(196N)\cos 25° = 53.3N$

$$W_f = (f_k\cos\phi)s = (53.3N)(\cos 180°)(3.80m) = -202J$$

(e) $W_{tot} = W_F + W_w + W_n + W_f = 532J - 315J + 0 - 202J = 15J$

(f) $W_{tot} = K_2 - K_1$, $K_1 = 0$, so $K_2 = W_{tot}$

$\frac{1}{2}mv_2^2 = W_{tot}$, so $v_2 = \sqrt{\frac{2W_{tot}}{m}} = \sqrt{\frac{2(15J)}{20.0kg}} = 1.2m/s$

6.68 (a) $W = \int_0^{x_2} F\,dx = \int_0^{x_2}(kx - bx^2 + cx^3)\,dx = \frac{k}{2}x_2^2 - \frac{b}{3}x_2^3 + \frac{c}{4}x_2^4$

$$W = (500N/m)x_2^2 - (233N/m^2)x_2^3 + (3000N/m^3)x_2^4$$

When $x_2 = 0.050m$, $W = 0.12J$

(b) When $x_2 = -0.050m$, $W = 0.17J$

(c) It's easier to stretch the spring, the quadratic term $-bx^2$ is always in the $-x$ direction, and so the needed force, and the needed work, will be less when $x_2 > 0$.

6.82 $W_{tot} = (6.00\,kg - (0.25)(8.00\,kg))(9.80\,m/s^2)(1.50\,m) = 58.8\,J$

This work increases the kinetic energy of both blocks

$W_{tot} = \frac{1}{2}(m_1 + m_2)v^2$, so $v = \sqrt{\frac{2W_{tot}}{m_1+m_2}} = \sqrt{\frac{2(58.8\,J)}{14.00\,kg}} = 2.90\,m/s$

6.93 (a) The volume $V = (7500\,L)\left(\frac{1\times10^{-3}\,m^3}{1\,L}\right) = 7.50\,m^3$

$m = \text{density} \times \text{volume} = (1.05\times10^3\,kg/m^3)(7.50\,m^3) = 7.875\times10^3\,kg$

Then $W = mgh = (7.875\times10^3\,kg)(9.80\,m/s^2)(1.63\,m) = 1.26\times10^5\,J$

(b) $P_{av} = \frac{\Delta W}{\Delta t} = \frac{1.26\times10^5\,J}{(24\,h)(3600\,s/h)} = 1.46\,W$

6.96 (a) Along this path y is constant. $y = 3.00\,m$

$W = 2y\int_{x_1}^{x_2} x\,dx = (2.50\,N/m^3)(3.00\,m)\frac{(2.00\,m)^2}{2} = 15.0\,J$

(b) Since the force has no y-component, no work is done moving in the y-directio

(c) Along this path, y varies with position along the path, $y = 1.5x$.

so $F_x = 2(1.5x)x = 1.5\,2x^2$

$W = \int_{x_1}^{x_2} F\,dx = 1.5\,2\int_{x_1}^{x_2} x^2\,dx = 1.5\,(2.50\,N/m^2)\frac{(2.00\,m)^3}{3} = 10.0\,J$

6.101 (a) Denote the position of a piece of the spring by l; $l=0$ is the fixed point and $l=L$ is the moving end of the spring. Then the velocity of the point corresponding to l, denoted by u, $u = v\frac{l}{L}$. The mass of a piece of length dl is $dm = \frac{M}{L}dl$.

so $dK = \frac{1}{2}(dm)u^2 = \frac{1}{2}\frac{Mv^2}{L^3}l^2\,dl$, and $K = \frac{Mv^2}{2L^3}\int_0^L l^2\,dl = \frac{Mv^2}{6}$

(b) $\frac{1}{2}kx^2 = \frac{1}{2}mv^2$, so $v = \sqrt{\frac{k}{m}}x = \sqrt{\frac{3200\,N/m}{0.053\,kg}}\times 2.50\times10^{-2}\,m = 6.1\,m/s$

(c) With the mass of the spring included, the work that the spring does goes into the kinetic of both the ball and the spring, so $\frac{1}{2}kx^2 = \frac{1}{2}mv^2 + \frac{1}{6}Mv^2$

$v = \sqrt{\frac{k}{m+\frac{M}{3}}}\,x = \sqrt{\frac{3200\,N/m}{0.053\,kg + (0.243\,kg)/3}}\times 2.50\times10^{-2}\,m = 3.9\,m/s$

(d) $\frac{1}{2}mv^2 = \frac{1}{2}\frac{kx^2}{(1+M/3m)} = 0.40\,J$ and $\frac{1}{6}Mv^2 = \frac{1}{2}\frac{kx^2}{(1+3m/M)} = 0.60\,J$